

Continuity and Rules of Differentiation

ECON 441: Introduction to Mathematical Economics
Lecture 9

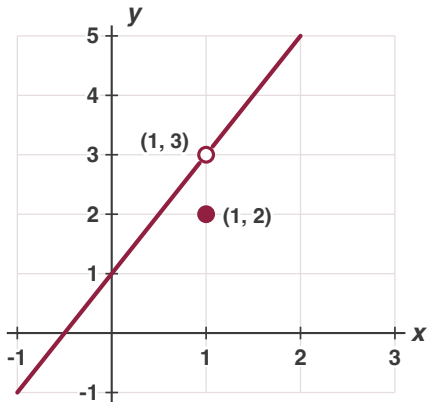
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Continuity of a Function

A function $y = f(x)$ is said to be **continuous** at a if $\lim_{x \rightarrow a} f(x)$ exists and

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Discontinuity: Example



$$y = \begin{cases} 2x + 1 & \text{if } x < 1 \\ 2 & \text{if } x = 1 \\ 2x + 1 & \text{if } x > 1 \end{cases}$$

In this example, the limit exists but the function is not continuous.

Differentiability and Continuity

$f'(x_0)$ exists if the following limit exists:

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

A function $y = f(x)$ is continuous at x_0 if

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

Any connection?

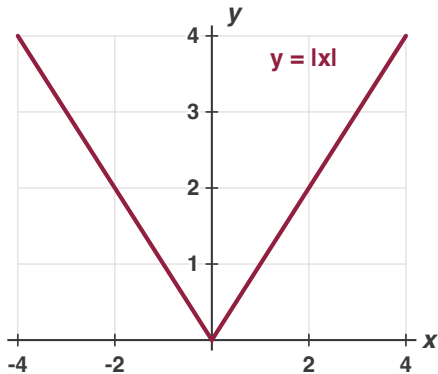
Differentiability and Continuity

Continuity is a necessary condition for differentiability, but it is not sufficient.

f is not continuous $\implies f$ is not differentiable

f is continuous $\implies f$ could be differentiable or not

Continuous but Not Differentiable



$$y = |x|$$

This function is continuous but not differentiable.

Note: $|x|$ represents the absolute value of x . For example, $|2| = 2$ and $|-2| = 2$.

So How to Differentiate Functions?

Rules of differentiation, easier than taking the limit each time

Constant function rule: for $f(x) = k$, $f'(x) = 0$

Power function rule: for $f(x) = x^n$, $f'(x) = nx^{n-1}$

Generalized power function rule: for $f(x) = cx^n$, $f'(x) = cnx^{n-1}$

Rules of Differentiation

Two or more functions of one variable

Sum-difference rule

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

Product rule

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

Rules of Differentiation

Two or more functions of one variable

Quotient rule

$$\frac{d}{dx} \frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

Inverse function rule

$$\frac{dx}{dy} = \frac{1}{dy/dx}$$

Rules of Differentiation

Functions of different variables

Chain rule: for $z = f(y)$, $y = g(x)$

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx} = f'(y)g'(x)$$

Homework Questions

- Exercise 7.1: 3
- Exercise 7.2: 3 (d), (e), 7, 8
- Exercise 7.3: 1-6

Textbook reference: 6.7, 7.1-7.3